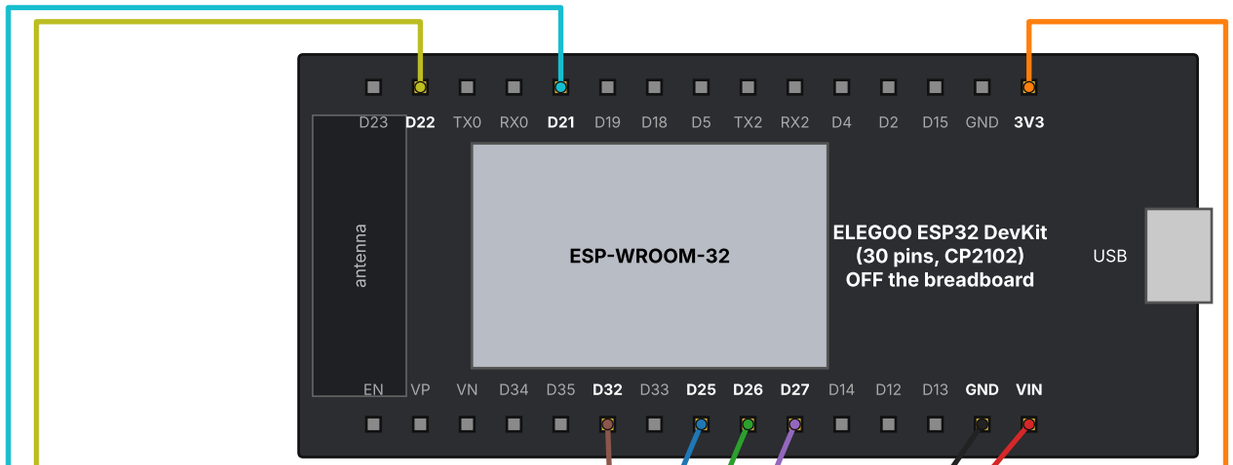
RoomCleaner winch station Rev D on a breadboard with an ELEGOO ESP32 DevKit (off the board, 9 female-to-male jumpers).
Same nets as the XIAO breadboard, the perfboard and the KiCad schematic.




```
ESP32 pin -> hole   (female end on the ESP32, male end in the board)
 D25 -> d6   STEP
 D26 -> d7   DIR
 D27 -> d8   EN
 D32 -> h8   HOME
 GND -> h6   GND
 VIN -> h5   VIN 5 V
 D21 -> d9   SDA
 D22 -> d10  SCL
 3V3 -> T+   3V3 (top rail, row 19)
```

Do not use D0/GPIO0, D2, D5, D12, D15 (boot strapping) or VP/VN/D34/D35 (input only, no pull-ups).
VIN gets 5 V from the buck through D1 and JP1: plug JP1 in ONLY with USB unplugged.

ENCODER SDA → a9, SCL → a10,
VCC → T+, GND → T- (AS5600 DIR pad → T-)

12 V is ONLY in row 20 (f-j) and the red wire j20 → j36. Both + rails are 3V3, both - rails are GND.

